

Detailed analysis on thermal, microstructural, mechanical, and morphological features of side chain liquid crystalline polymer/isotactic polypropylene graft copolymers: Effect of grafted and ungrafted polymer units

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Abstract

The main scope of this comprehensive study is to investigate the effects of poly(p-benzophenoneoxycarbonylphenyl acrylate), poly(BPOCPA), which presenting as only graft units or both graft and ungrafted units in the matrix, on the fundamental features of isotactic polypropylene (IPP). The graft copolymerization of BPOCPA onto IPP was performed with the aid of bulk melt polymerization at varying monomer content levels ranging from 5% to 40%. The thermal behavior, crystal quality, mechanical performance, and surface morphology of the samples were investigated by means of differential scanning calorimeter, X-ray diffractometer (XRD), universal mechanical test, and scanning electron microscope (SEM) techniques. Thermal analyses depicted that there existed the noteworthy enhancements in both crystalline melting temperatures and percent crystallinities of matrix polymers. Furthermore, according to XRD results, a and b parameters increased significantly at low percentages of the graft units, while the parameter c decreased in all products in consistence with the content. As for the mechanical characterization, the grafting led to remarkable improvements in modulus, tensile and impact strength of the products. SEM micrographs indicated that the samples were completely homogeneous without any phase separation and the products exhibited brittle nature with some ductility.

KEYWORDS

copolymers, grafting, mechanical properties, structure–property relationships, thermal properties

1 | INTRODUCTION

Polypropylene (PP) is well-known and cheap thermoplastic polymer besides has good reputation due to its usage in the varying application fields such as packaging, laboratory equipments, textiles, automotive components, and so forth.^{1,2} Among the many types of PP, depending on the tacticity, isotactic propylene (IPP) had prior

characteristics features such as relatively more ordered structure, higher melting point, and larger degree of crystallinity, which makes it more preferable to be imparted the desired superior features with the physical or chemical modification. Despite of these charming properties, there existed some unfavorable sides of IPP such as brittleness at low temperatures, low-impact toughness lack of reactive site, low-mechanical endurance, low

-hydrophobic character, sticking problem, photo oxidation sensitivity, dye uptake, and so forth.³ Thus, the researchers have focused on the improving of these properties of the IPP in order to get rid of these drawbacks with the aid of the certain modifications techniques.^{4–6} Therein, especially, the graft copolymerization method plays a significant role due to the enabling to modify the grafted units by changing the functional groups in their molecular backbone. Moreover, it should be pointed out that, via the graft copolymerization, the chemical bond is formed between the components without any the phase separation that makes worse the polymers since the wholly miscible system illustrates the better mechanical property comparing to the partially miscible or immiscible systems.⁷ In addition, the formation of the better compatibility between the components brings about the reduction in interfacial energy, good dispersion, inhibition effect of the segregation and enhanced adhesions,^{8,9} which means that the polymer shows the comparatively better mechanical performance. Furthermore, the grafted units in the polymers could lead to increase in the degree of freedom of the main chains by enhancing the free volume features of the matrix.

In addition to that, it is known that the groups presenting in the grafted units affect highly the crucial characteristics of the thermoplastics such as IPP.^{10,11} Namely, liquid crystalline polymer (LCPs) with their remarkably rigid, rod-like, and stiff molecular backbone causes to form well-ordered structure and better arrangement with dimensional stability in the polymer matrix by forming highly orientations and alignments. Moreover, the existence of LCPs in the polymer matrix, which gains the ability to show the anisotropic behavior due to the underlying or subunit groups called as mesogens, leads to significant advance in the mechanical performance by means of their comparatively higher load capacity.¹² This reinforcement effect is caused by that outstanding modulus (4–20 GPa) and the strength (139–213 MPa) of the injection-molded LCP substantially larger than that neat conventional thermoplastic had. Furthermore, the presence of LCPs phase dispersed homogeneously in the polymer matrix gives rise to occurrence of the fibrillar manner creating the fiber-reinforcement effect in the thermoplastic polymers.¹³ Namely, the LCP phase in the polymer blends provides the formation the fibrillar structure throughout uniaxial extension flow and hence, gives rise to reinforcement of the polymer matrix with exhibiting higher mechanical performance. In addition, the another advantage regarding the usage of LCPs is that they reduce the melt viscosity of the blends during the processing since they possess lower melt viscosity compared to thermoplastics.¹⁴ In the light of that, in our previous study, the reinforcement of high-density

polyethylene (HDPE) was performed with the side chain LCP by graft copolymerization. The remarkably increment in the crystalline melting temperature of HDPE, tensile strength (38% increase), Young's modulus (67% increase) as well as the considerable expansion in the unit cell parameters of HDPE orthorhombic structure were obtained thanks to advances in the orientation and alignment of HDPE chains in the polymer matrix. Furthermore, Li et al. tried to improve the physical properties such as mechanical and heat resistant properties of glass fiber-reinforced polypropylene (GFRPP) by using LCP. They found that the yield and the impact strength increased by 62.7% and 18.1% due to both the existence of LCP and the high-quality interfacial adhesion in the polymer matrix, respectively.¹⁵ Moreover, as for LCP/PP blends, the tertiary components presenting in the polymer matrix may create the positive effects in such blends. Namely, the rheological and morphological properties of the PP/LCP blends depending on the silica addition were investigated by Foudazi and Nazockdast.¹⁶ According to the obtained experimental results, the used hydrophobic silica for PP/LCP blend had a good compatibilization capability, which causes to achieve the better properties. Accordingly, the addition of the nanosilica particles as filler to LCP/PP blends created the improvement effects in the mechanical performance with the formation of LCP fibrillation and the blends composed of LCP/PP/SiO₂ depicted comparatively higher tensile strength and modulus.^{17,18} In order to improve the compatibility, the tertiary compatibilizers agent was also utilized in the LCP/PP blends.¹⁹ Moreover, the advanced mechanical endurance and easier processability for PP were obtained by means of the usage of LCP increasing the mean diameter of the fibrils in the polymer matrix.²⁰ On the other hand, it could be emphasized that poly(p-benzophenoneoxycarbonylphenyl acrylate), poly(BPOCPA) bearing the pendant aromatic mesogenic groups in the polymeric backbone possessed the mesophase properties of according to the papers announced previously.^{21,22} They founded that poly(BPOCPA) exhibited liquid crystalline behavior by forming the glassy nematic order phase observed at the range from 144 to 199°C.

Although IPP composites and blends were studied in many studies, there did not exist any research delving into how the presence of only grafted units or both grafted and ungrafted units in the polymer matrix affects thermal, microstructural, mechanical, and morphological properties of IPP. In order to fill this gap, we tried to improve the crucial characteristic peculiarities of IPP with the aid of poly(BPOCPA) announced as the side chain LCP. By considering poly(BPOCPA) molecules possessing well-ordered and regular structure due to their glassy nematic phase, the graft copolymers containing the different percentage of poly(BPOCPA) were manufactured. The existence of poly(BPOCPA) as grafted or

ungrafted in the matrix probably gained better orientation and advanced alignment ability to IPP chains, which potentially resulted in the remarkable mechanical reinforcements, enhancement in the quality of crystal units, increment in the crystalline melting temperature as well as the formation of more durable morphological appearance with fibrillar structure.

2 | EXPERIMENTAL

2.1 | Materials

Acryloyl chloride (AC), 4-hydroxybenzophenone (HBP), and sodium hydroxide (NaOH) were purchased from Alfa Aesar A.G., whereas p-hydroxybenzoic acid (HBA) and thionyl chloride were supplied from Merck A.G. These analytical grade main chemicals for the synthesis of BPOCPA were used without purification. The initiator, dicumyl peroxide (DCP, Merck A.G) was utilized for the homopolymerization and graft copolymerization reactions. The solvents, which used for the conducting of the some experimental processes and washing the products, xylene, acetone, dimethyl sulfoxide (DMSO), dimethyl formamide (DMF), ethanol, and methanol were in analytical grade and purchased from Merck A.G., except for ethanol, which was the technical grade. Isotactic polypropylene (IPP) was supplied in the granule form by Turkish Petrochemical Industry (PETKIM) under the trade name MH 418. IPP granules were dissolved in hot xylene (138–139°C) by refluxing throughout 2 days and then, precipitated with sufficient amount of ethanol. Then, the obtained precipitates were filtered with the usage of the Buchner funnel, dried under vacuum under 50°C for 5–6 h. The dried IPP particles were grounded by hand grinding by cooling in liquid nitrogen in order to obtain fine and powder form IPP having roughly close particle size.

2.2 | Synthesis of the monomer, BPOCPA

The monomer, BPOCPA was synthesized in the three steps; (I) p-acryloyloxybenzoic acid, ABA, (II) acryloyloxybenzoyl chloride, ABC and (III) the monomer, BPOCPA as reported in our previous study.²² At first step, 69.0 g of HBA (0.5 mol) were added to 0.5 M of NaOH solution and the prepared solution were cooled at the temperature of 0–5°C. Then, equivalent mol of AC (0.5 mol) was added to this solution drop by drop by stirring about 1 h. The solution was precipitated with the addition of diluted HCl, then filtered and washed excessively with water to get rid of

the formed salts. The obtained intermediated product, ABA was purified 3–4 times via the recrystallization techniques from acetone. The reaction yield was recorded for ABA as 75 wt%. At second step, ABA was refluxed approximately 7–8 h in thionyl chloride containing trace amount of dimethyl formamide to prepare ABC. The reaction yield of ABC was found to be about 98.0%. Finally, equimolar solution of ABC and HBP in xylene containing pyridine was stirred for 3–4 days in order to synthesize BPOCPA. During the reaction, dark brown residue was obtained. For the separation of this impurity, the mixture was heated and the solution containing only monomer was separated by decantation. The monomer was recrystallized in xylene for purification and obtained with the yield of 60%. Therein, it should be pointed out that at all steps, all the products were characterized by means of conventional experimental techniques such as FT-IR, differential scanning calorimeter (DSC), ¹H-NMR. It was deduced that all the prepared compounds showed good correlation with the results obtained from our previous study.²³

2.3 | Preparation of homo and graft products of BPOCPA

The homopolymerization reaction of the monomer, BPOCPA was carried out by both bulk melt and solution polymerization methods at the reaction conditions described previously.²³ The graft copolymerization of BPOCPA onto IPP was also performed via only bulk polymerization method. The six different mixtures (as two series) composed of powder IPP (9.5, 9.0, 8.5, 8.0, 7.0, and 6.0 g), BPOCPA (0.5, 1.0, 1.5, 2.0, 3.0, and 4.0 g) as well as 2% of DCP (depending on the weight of BPOCPA) were prepared in a mortar by grinding with hand extensively for 1 h. These mixtures put into the vacuumed glass reaction ampoules, then heated up to 160°C by keeping the temperature constant for 1 h. After the reactions were completed, the obtained products were firstly washed with acetone (250 ml × 4) to remove monomer residuals and undesired impurity. Herein, one of the products series were washed efficiently with hot DMSO (at about 130–140°C) to remove all the ungrafted poly(BPOCPA) and dried in vacuum at 40°C, hence these products were called throughout this current study as “copolymer” containing only grafted poly(BPOCPA). The other products series were not washed with hot DMSO, thus they were defined as “coproducts” including not only poly(BPOCPA) homopolymer molecules but also poly(BPOCPA)-g-IPP graft units. The gravimetrically calculated percent grafting findings were given as the averages of the data obtained from minimum three

experiments. The schematically representation concerning the grafting of BPOCPA onto IPP was given in Figure 1.

2.4 | Instruments

FT-IR analyses of the BPOCPA, all synthesized intermediate molecules, IPP, poly(BPOCPA) and the obtained graft products were performed by means of Shimadzu 8400-S FT-IR spectrophotometer in the range of $4000\text{--}400\text{ cm}^{-1}$ at the transmittance mode. The melting and crystallization behavior of neat IPP and the obtained products were evaluated by Shimadzu DSC 60 DSC in the temperatures ranging from 30 to 300°C with the heating rate of $10^\circ\text{C}/\text{min}$ under nitrogen atmosphere. The microstructural properties of the products were determined by Rigaku Multiflex X-ray diffractometer at room temperature. Cu-K α monochromatic beam was used as a radiation source and the wavelength was $1.54\text{ \AA}$. The diffraction patterns of products were recorded in the 2θ range of $10^\circ\text{--}60^\circ$, a scan rate of $5^\circ/\text{min}$ and a scanning step of 0.02° . The tensile test samples in the dumb-bell shape (a gage length of 50 mm , a thickness of 2 mm , and a width of 7.6 mm) were prepared by Daga Instruments Micro-injector. The injection processes were conducted at the temperature of 220°C and at the pressure of 8 bar . The tensile test measurements were done by using LLYOD LR5K Mechanical Tester at room temperature with load cell of 5 kN at crosshead speed of $50\text{ mm}/\text{min}$ according to ASTM D-638 standard. The stress versus strain graphs, tensile strength, and moduli for the test samples were obtained directly from the software of the instrument. Coesfeld Material Test Pendulum Impact

Tester was used to determine the impact strengths of unnotched neat IPP and the product samples with dimension of 1 mm (thickness) $\times$ 7 mm (width) at room temperature. At least four specimens for each sample were tested and the results were given as averages of these tested samples. In order to examine the morphology of the tensile and impact fractured surfaces of samples, JEOL 6390-LV Scanning electron microscope (SEM) operated with an accelerating voltage of 20 kV and with a resolution power of 3 nm were used. Prior to the SEM analysis, The fractured surfaces obtained from the samples after the tensile and impact tests were located on carbon conductive adhesive tape (known as carbon sticker) and then, the samples were sputter-coated with Au since the samples had insulating characters.

3 | RESULTS AND DISCUSSION

3.1 | Graft copolymerization of BPOCPA onto IPP

The percentage of poly(BPOCPA) presenting in the copolymers containing only grafted units and coproducts including both grafted and ungrafted polymers were determined by the gravimetric calculations, and the obtained results were tabulated numerically in Table 1, describing how the percent poly(BPOCPA) in the copolymers and coproducts as well as percent grafting changed with BPOCPA content levels. Prior to discussion on the graft copolymerization, it was to be emphasized that the copolymers had good purity since the washing process with excessive amount of DMSO was carried out efficiently with several times for the samples. As the

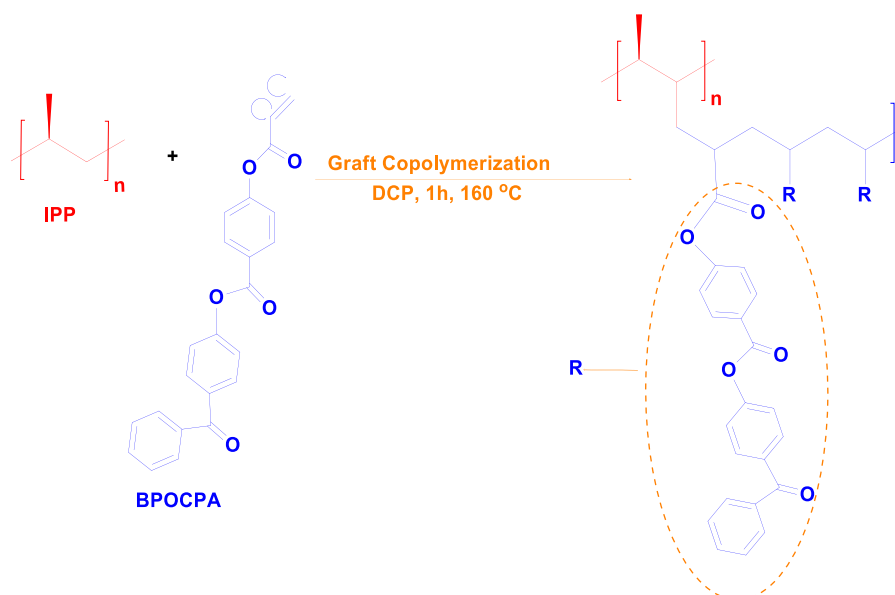
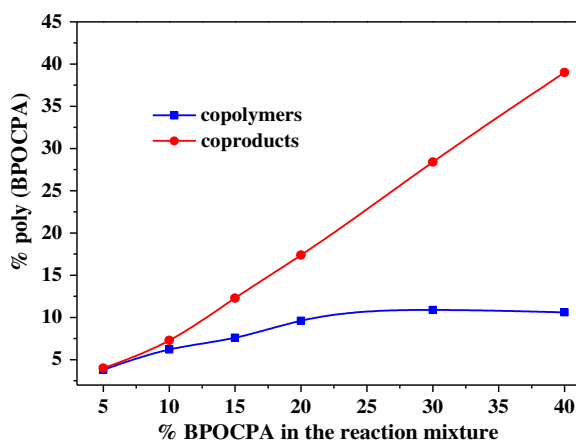


FIGURE 1 The simplified possible graft copolymerization reactions of BPOCPA onto IPP. BPOCPA, p-benzophenoneoxycarbonylphenyl acrylate; DCP, dicumyl peroxide; IPP, isotactic polypropylene [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 The dependence of poly(BPOCPA) content in the products on the BPOCPA percentage in the reaction mixture

% IPP in reaction mixture	95	90	85	80	70	60
% BPOCPA in reaction mixture	5	10	15	20	30	40
% poly(BPOCPA) in the coproducts	4.0	7.3	12.3	17.4	28.4	39.0
% poly(BPOCPA) in the copolymers	3.8	6.2	7.6	9.6	10.9	10.6
% ungrafted poly(BPOCPA)	0.2	1.1	4.7	7.8	17.5	28.4
% Grafting	76.8	63.7	50.0	48.4	32.0	15.9

Abbreviations: IPP, isotactic polypropylene; poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate).

**FIGURE 2** The variation of poly(BPOCPA) content in the products with the BPOCPA percentage in the reaction mixture. IPP, isotactic polypropylene; poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate) [Color figure can be viewed at wileyonlinelibrary.com]

monomer concentration increased in the reaction mixtures, the percentage of poly(BPOCPA) in both the copolymers and the coproducts increased almost linearly as seen in Figure 2, however the extents of grafting were not proportional with the conversions, and the grafting yield diminished drastically with the monomer content. Moreover, the results showed that the extent of grafting reached 9.6, 10.9, and 10.6% poly(BPOCPA), forming almost a plateau at about 10% poly(BPOCPA), with 20%, 30% and 40% BPOCPA, respectively. Furthermore, the maximum percent grafting was found to be 76.8% at 5% BPOCPA, while the minimum values was recorded as 15.9% at 40% BPOCPA. These results revealed that the grafting took place via the radicals forming on IPP chains in majority, rather than the reactions between propagating poly(BPOCPA) radicals and IPP chains. Namely, IPP had comparatively more tertiary hydrogens which were available for hydrogen abstraction, which was less energetically favored for poly(BPOCPA) compared to IPP, as a consequence of that, the probability of the formation of β scission reactions may increase instead of grafting onto the polymer.²⁴ It can be concluded that the higher

probability of radical formation on IPP chains at relatively lower monomer concentrations essentially resulted in higher grafting efficiencies on IPP chains. Steric hindrance created by crowded and polar side groups of poly(BPOCPA) units, on the other hand, were presumed to account for the observed decreases in the grafting yields.

The molecular characterization of neat IPP, poly(BPOCPA), the copolymer with 10.6% BPOCPA and the coproducts including 39.0% BPOCPA were done with the aid of FTIR analysis. The obtained spectrums with the indications of characteristic bands were exhibited in Figure 3. As for IPP, the spectrum indicated the identical bands of stretching vibrations of CH_3 group at 2956 cm^{-1} and those of CH_2 group at 2924 and 2852 cm^{-1} . The bands due to bending vibrations of CH_3 and CH_2 groups were observed at 1377 and 1458 cm^{-1} , respectively. Moreover, FTIR spectrum of homopolymer, poly(BPOCPA) showed the characteristic band at 1741 cm^{-1} corresponding to $\text{C}=\text{O}$ stretching vibrations of the ester groups. The strong band also detected at about 1651 cm^{-1} was due to $\text{C}=\text{O}$ stretching vibrations of the benzophenone group. Aromatic $\text{C}=\text{C}$ stretching vibrations were observed at 1502 and 1597 cm^{-1} . The strong bands at 1197 and 1267 cm^{-1} were due to esteric $\text{C}-\text{O}-\text{C}$ stretching vibrations. The bands seen between 1000 and 1200 cm^{-1} were attributed to mono-substituted aromatic $\text{C}-\text{H}$ bending vibrations. It was obvious that any bands attributed to $\text{C}-\text{H}$ out-of-plane bending vibrations due to vinylic groups were not detected in the spectrum of poly(BPOCPA). This result confirmed clearly that the polymerization reaction of BPOCPA took place through the vinylic group. In addition, as seen apparently from the spectrums, all the characteristic peaks concerning both neat IPP and poly(BPOCPA) molecules also observed in the FTIR spectra belonging to both the copolymer with 10.6% BPOCPA and the coproducts including 39.0% BPOCPA. This might be seen as an apparent sign regarding to taking place of the graft copolymerization of BPOCPA on IPP via chemically bonding. Furthermore, since the percentage of poly(BPOCPA) in the coproducts were relatively larger than that the copolymers had, it

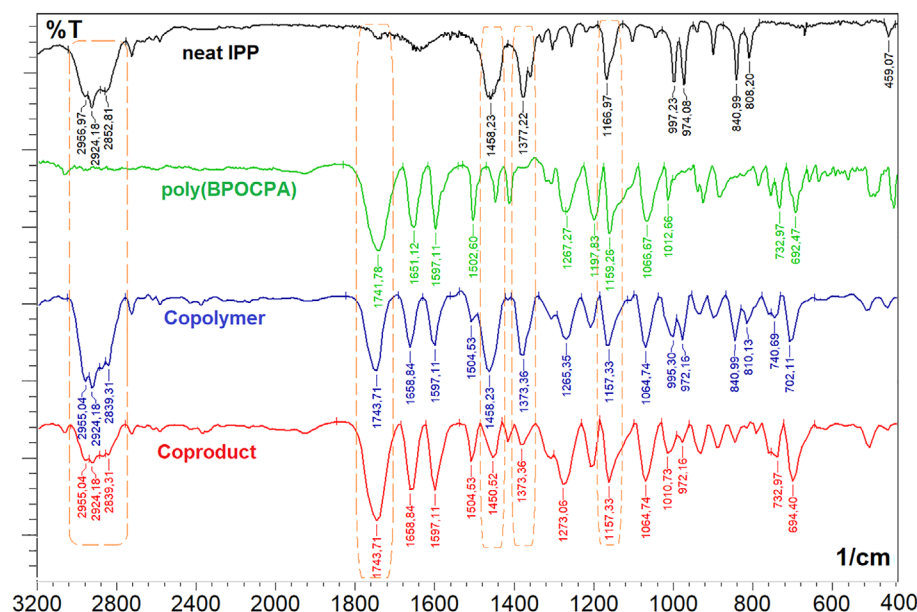


FIGURE 3 FTIR spectrums of neat IPP, poly(BPOCPA), the copolymer with 10.6% of grafted poly(BPOCPA) and the coproducts including 39.0% poly(BPOCPA). IPP, isotactic polypropylene; poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate) [Color figure can be viewed at wileyonlinelibrary.com]

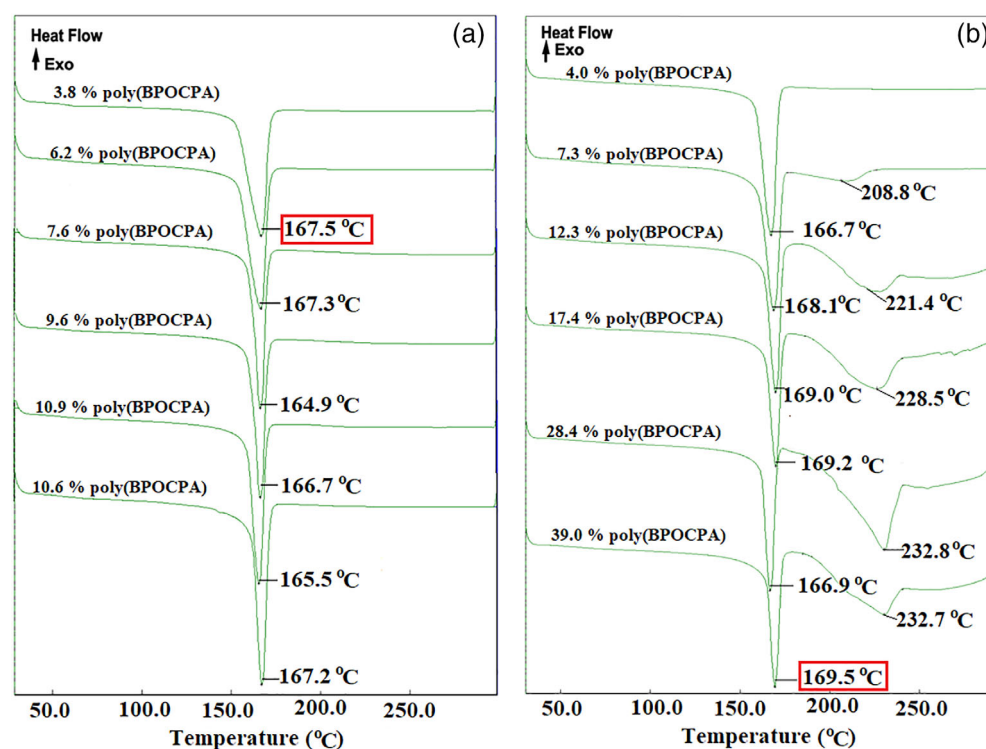


FIGURE 4 The DSC thermograms of (a) the copolymers and (b) the coproducts including varying percentage of poly(BPOCPA). DSC, differential scanning calorimeter; poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate) [Color figure can be viewed at wileyonlinelibrary.com]

was observed obviously from the spectrum that the characteristics bands of poly(BPOCPA) were comparatively more dominant.

3.2 | Melting and crystallization behavior of the products

The role of grafted and ungrafted poly(BPOCPA) polymers presenting in the matrices on the thermal properties

of IPP were examined by DSC analyses. The obtained DSC thermograms belonging to the copolymer and the coproducts with varying amount of poly(BPOCPA) were depicted in Figure 4. It was deduced from the figure that the endothermic heat flows observed in the ranging from 164 $^{\circ}\text{C}$ to 170 $^{\circ}\text{C}$ were assigned to the crystalline melting temperature of IPP domains. Since the melting temperature of the homopolymer was reported as 231 $^{\circ}\text{C}$,²⁵ as expected, the endotherms ascribed to the crystalline melting of the poly(BPOCPA) ungrafted units were recorded

in the range of 208.8–232.8°C in the thermograms of all the coproducts. Furthermore, one can see easily from Figure 4(b) that the relative endotherms became more dominant with the increasing of the poly(BPOCPA) level in the matrices. Moreover, it was apparent that the graft copolymerization led to remarkable enhancement in the melting temperature of IPP. The maxima in the melting points were found to be 167.5°C and 169.5°C at the copolymer with 3.8% poly(BPOCPA) and the coproducts with 39.0% poly(BPOCPA), respectively.

In the melting behavior of the products presented, a combination of two opposing factors contributing simultaneously may be effective within the range of concentrations studied. The LCP phase stimulating a range of crystalline structures (heterogeneous phase nucleation) of different sizes and shapes of the IPP may lead to the breakdown of the crystal structure of IPP and the consequent behavior in the melting temperature can be observed. The broadening of the softening range may possibly be due to a range of widely different sizes of the crystal structures, which increase by increasing proportions of LCP. Thus, it may be assumed that the effect of crystallinity in the ultimate blends remains almost identical to that of the virgin PP, which, in effect, might have compensated the high-melting point of stereoregular isotactic PP.²⁶ Regarding this explanation, heterogeneous phase nucleation might occur due to grafting of the side chain LCPs, and different sizes of crystalline structures might be effective, leading to dispersed melting temperatures compared to IPP. Also, this heterogeneous nucleation might have given rise to changes in crystallinity of products. According to the crystallization theory, the process of crystallization takes place in two stages: nucleation of individual crystals and their subsequent growth. Each stage occurs at a characteristic rate and both the processes contribute to the overall rate of crystallization, considered as the mass of the crystalline material formed from the melt in unit time. The rate of the nucleation stage determines the number of nuclei, which arise per unit mass per unit time, while the rate at which the crystal growth front progresses in units of velocity, accounts for growth. The faster the nucleation process, the higher will be the nucleation density and hence the average diameter of spherulites will become smaller. Nuclei smaller than the critical nucleus are continuously formed and disrupted in the melt, whereas those of critical dimensions are able to grow. Depending on the chemical composition of the nuclei, the nucleation may be homogeneous or heterogeneous. In the first case, the atoms which aggregate to form a nucleus arise from the polymer chains themselves, while an extraneous nucleating particle (such as impurities, catalyst residues, chain end groups, etc.) act as a nucleus in the heterogeneous

nucleation type. In order that foreign particles are able to act as nucleating agents, the interfacial energy between their surface and the polymer crystal must be very low.

In addition to that, DSC thermograms of the poly(BPOCPA)-g-IPP copolymers produced upon changing DCP concentration for constant mixing ratio of BPOCPA in the reaction mixture (30%) were shown in Figure 5. While single melting endotherms were seen in the thermograms of the products obtained by using 2% DCP, and by using 1% DCP, double melting endotherms were observed for the samples when initiator concentration was increased from 1% to 3–4% in the productions. The first melting peak, seen as shoulder, was at about 163°C, and the second one was recorded at about 170°C. This thermal behavior was directly related to the difference in the crystallographic symmetries of IPP such as α , β , and γ crystalline forms. Namely, all the forms are constituted from chains in a_31 helical conformation with a common repeat distance (approximately 6.5 Å), but differentiate in unit cell symmetry, structural disorder and interchain packaging. The α -form is arranged on a “monoclinic unit cell” and characterized by the changes in b-axis direct, of layers parallel to the ac-plane and composed of left handed (L) or right-handed (R) helices. The position of methyl groups in IPP are at down or up sides, and the changes in the position affect α -phase of polypropylene, leading to α_1 or α_2 modification. Two limiting structures were postulated for the α form: a disordered limiting structure (α_1) with the corresponding

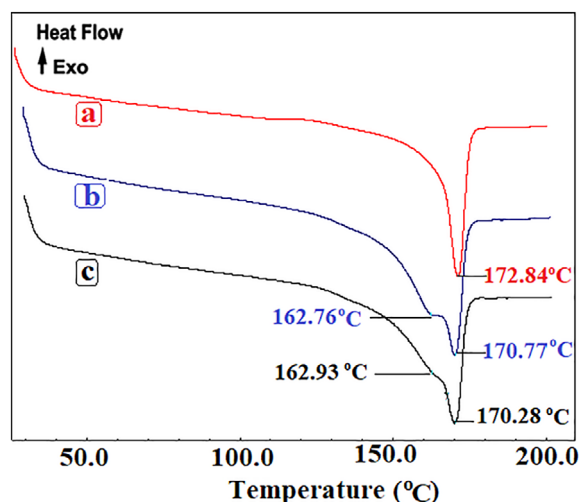


FIGURE 5 DSC thermograms, obtained of the poly(BPOCPA)-g-IPP copolymers produced by using the initiator (a) 1% DCP, (b) 3% DCP, and (c) 4% DCP. DCP, dicumyl peroxide; DSC, differential scanning calorimeter; IPP, isotactic polypropylene; poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate) [Color figure can be viewed at wileyonlinelibrary.com]

crystallographic symmetry being $C2/c$ and an ordered limiting structure (α_2) having crystallographic symmetry being $P2_1/c$.²⁸ Reported that IPP may exhibit two melting endotherms in DSC, even if the sample is composed of only the α form. The first peak of two melting endotherms observed at low-heating rates on the samples of isotactic PP (α form) was attributed to the melting of crystals having a structure with α_1 modification and the second one was related to the melting of crystals with a structure nearer to α_2 modification. Accordingly, in the thermograms of the samples produced by using 3% and 4% DCP, the first peak observed as shoulder at about 163°C probably denoted the melting of the crystals with α_1 modification. The second peak recorded at about 170°C presumably represented the melting of the crystals with α_2 modification.

In order to determine the effect of poly(BPOCPA) presenting as grafted and ungrafted units on the degree of crystallinities of all the products, the percent crystallinities for the each samples were calculated by using the heat of fusion (ΔH_f°) value taken from directly the melting endotherms by using software of the instruments and the equation was given below:

$$X_c(\%) = \frac{\Delta H_f}{\Delta H_f^\circ} \times 100.$$

With respect to this equation, the heat of fusion of 100% crystalline IPP (ΔH_f°) was assumed as 209 J/g.²⁷ The variations in percent crystallinity as a function of poly(BPOCPA) content in both copolymers and coproducts were plotted in Figure 6. The DSC analyses indicated that percent crystallinity in the IPP matrices increased with poly(BPOCPA) content in the copolymers at low percentages and reached the maximum 64.3% crystallinity (IPP used in the studies had 43.3% crystallinity) with the copolymer involving 9.6% poly(BPOCPA). The crystallinity then decreased at further contents. On the other hand, the presence of homopolymer molecules in the coproducts led to slightly smaller or comparable crystallinities in the products, except for the coproduct with 57% crystallinity and comprising 39% poly(BPOCPA). It could be hypothesized that the constitution of the reported side chain LCP poly(BPOCPA) molecules as graft units on IPP chains, with such regularly organized glassy nematic structure²¹ presumably led to the increases in the crystallinities at low contents up to 9.6% poly(BPOCPA). The regularly organized poly(BPOCPA) units might have acted as nucleating agents, conducting to more ordered packing of IPP chains in crystallites and resulting in the high crystallinities. At relatively high contents, the extensive nucleating due to increases in the content of the graft units, which might impede the growth of the crystals, and polar interactions between the

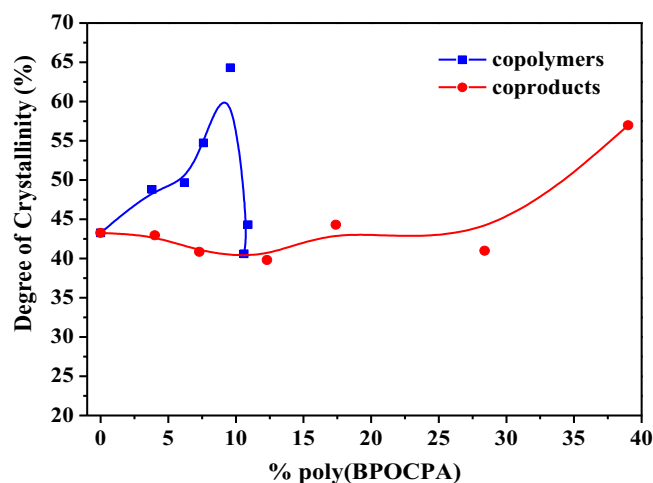


FIGURE 6 The variation of percent crystallinity with poly(BPOCPA) content in the copolymers and coproducts. poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate) [Color figure can be viewed at wileyonlinelibrary.com]

graft units probably prevented the increases in the crystallinities, which also account for the lower crystallinities in the coproducts.

3.3 | Microstructural behavior of the products

In order to find out the effect of the graft copolymerization on the microstructural behavior of the IPP matrices, in conjunction with their thermal behaviors, the unit cell parameters of crystal domains of the products were also estimated from the XRD patterns. The diffraction patterns of the copolymers and coproducts with the Bragg's angles (2θ) ranging from 10° to 40°, characteristic of α form, were presented in Figure 7. It was obvious from the figure that the IPP used in the graft copolymerization experiments was in the α form (monoclinic arrangement) without any formation of β and γ crystalline forms. Accordingly, the X-ray diffraction patterns of pure polypropylene displayed four characteristic peaks at 2θ of 14.14°, 17.12°, 18.6°, and 21.84°, which are assigned to the $(110)_\alpha$, $(040)_\alpha$, $(130)_\alpha$, $(111)_\alpha$, $(041)_\alpha$ reflections of the monoclinic subcell in which all the axes (a , b , and c) are different lengths,²⁸ Figure 7(a). Furthermore, it can be said that this confirmed the development of only α crystal in used IPP and also, α crystal arrangement had been conserved in the products throughout the graft copolymerization experiments as seen apparently in Figure 7. In addition, it was to be emphasized that the existence of poly(BPOCPA) polymers in the matrix brought about the some shifts toward to the right, broadening of the characteristic planes as well as decrement in the sharpness,

FIGURE 7 The XRD patterns of neat IPP, (a) the copolymers and (b) the coproducts with varying compositions of poly(BPOCPA). IPP, isotactic polypropylene; poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate); XRD, X-ray diffractometer [Color figure can be viewed at wileyonlinelibrary.com]

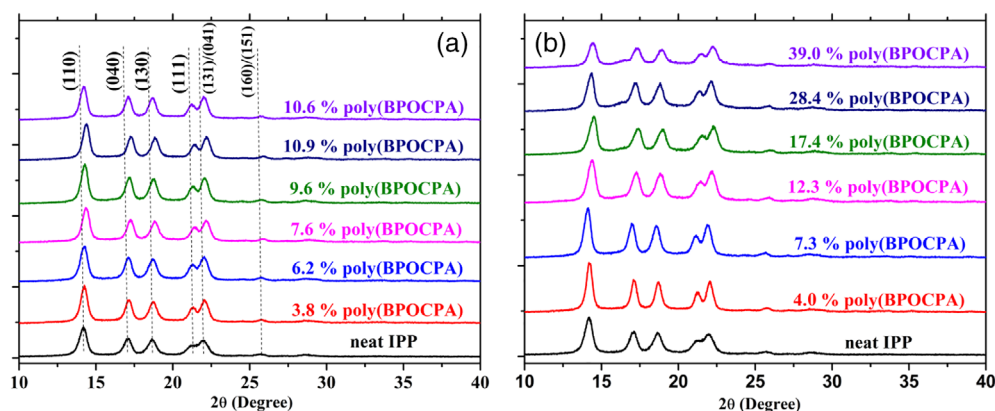


TABLE 2 The dependence of the unit cell parameters a, b and c, and crystal size in IPP matrix on poly(BPOCPA) content in the copolymer and copolymers

Samples	a (Å)	b (Å)	c (Å)	Crystal size (nm)
Neat IPP	6.632 ± 0.028	20.748 ± 0.026	6.549 ± 0.003	17.62 ± 0.10
3.8% poly(BPOCPA)	6.638 ± 0.008	20.741 ± 0.011	6.509 ± 0.003	20.45 ± 0.36
6.2% poly(BPOCPA)	6.674 ± 0.038	20.810 ± 0.039	6.510 ± 0.002	20.66 ± 0.03
7.6% poly(BPOCPA)	6.615 ± 0.008	20.706 ± 0.008	6.492 ± 0.007	22.45 ± 0.72
9.6% poly(BPOCPA)	6.605 ± 0.033	20.702 ± 0.032	6.422 ± 0.006	21.52 ± 0.26
10.9% poly(BPOCPA)	6.565 ± 0.002	20.572 ± 0.020	6.401 ± 0.005	19.60 ± 0.02
10.6% poly(BPOCPA)	6.603 ± 0.027	20.698 ± 0.035	6.419 ± 0.019	20.33 ± 0.56
4.0% poly(BPOCPA)	6.666 ± 0.023	20.800 ± 0.025	6.447 ± 0.002	22.90 ± 1.00
7.3% poly(BPOCPA)	6.685 ± 0.014	20.844 ± 0.023	6.443 ± 0.006	23.77 ± 1.65
12.3% poly(BPOCPA)	6.652 ± 0.016	20.788 ± 0.034	6.431 ± 0.008	23.23 ± 1.95
17.4% poly(BPOCPA)	6.645 ± 0.019	20.792 ± 0.022	6.424 ± 0.007	21.62 ± 1.47
28.4% poly(BPOCPA)	6.634 ± 0.007	20.790 ± 0.008	6.402 ± 0.001	19.46 ± 1.94
39.0% poly(BPOCPA)	6.624 ± 0.017	20.780 ± 0.028	6.393 ± 0.003	18.61 ± 1.32

Abbreviations: IPP, isotactic polypropylene; poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate).

Figure 7(b). Especially, in the coproduct samples, as percentage of poly(BPOCPA) increased in the polymer matrix, it was observed that these appearance changes became more noticeable, which may be a sign concerning the formation of some microstructural disorders, defects, and lattice distortions in the crystal structures of IPP.²⁹

On the other hand, both the crystal parameters and grain size (crystal size) belonging to all the products were calculated with regard to the monoclinic unit cell structures on the basis of a least square method using (*h k l*) planes and *d* values as announced in our previous study.²² The obtained results from computations given numerically in Table 2 and the dependence of the unit cell parameters a, b, and c, and the crystal sizes on the percentage of poly(BPOCPA) in the products were drawn in detail in Figure 8. The results revealed that the lateral dimensions of the monoclinic unit cell belonging to the

copolymer and coproducts were found to be considerably affected by poly(BPOCPA) content in the products. First, it can be stated that the parameters a and b lay on the values higher in the coproducts comparing to the copolymers at all contents of poly(BPOCPA). Both parameters increased significantly at low percentages of poly(BPOCPA) in both classes of the products. The maximum 6.685 Å was achieved with 7.3% poly(BPOCPA) in the coproducts (0.80% rise comparing to virgin IPP). In the copolymers the maximum 6.674 Å was recorded with 6.2% poly(BPOCPA) (0.63% increase). The maxima were then followed by decreases with the content, gradually and slowly in coproducts, but sharply in the copolymers. In addition, in spite of the decreases in the parameter a after the maxima, it remained still higher in the coproducts comparing to virgin IPP except for the coproduct involving 39% poly(BPOCPA). In the copolymers, on the

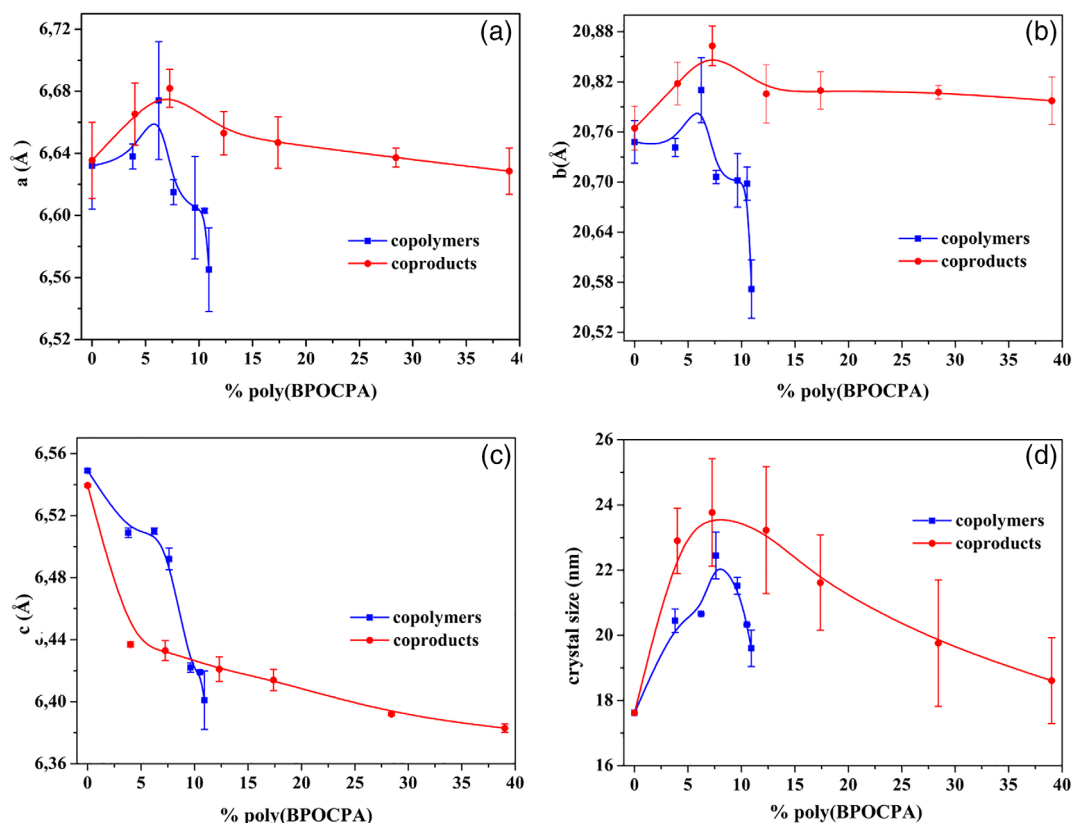


FIGURE 8 The variation of cell parameters (a) a, (b) b, (c) c and (d) crystal sizes of the both copolymer and coproducts with the poly(BPOCPA) percentage, calculated from the XRD patterns by using 110, 040, and 041 reflections. poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate); XRD, X-ray diffractometer [Color figure can be viewed at wileyonlinelibrary.com]

other hand, the decrease continued dramatically with the content, down to 6.565 Å seen at 10.9% poly(BPOCPA) (1.0% decrease), in Figure 8(a). A similar behavior was also observed in the parameter b except for that, after the maximum 20.844 Å (0.46% rise), also recorded with 7.3% poly(BPOCPA), it remained almost stationary by forming a plateau at about 20.790 Å between 12.3% and 39.0% poly(BPOCPA) in the coproducts as shown in Figure 8(b). It is assumed that the IPP chains in the unit cells were probably forced apart laterally by the polar graft units, thus giving rise to expansions in the lateral dimensions at low percentages of poly(BPOCPA). The weak interactions between polar side groups of the grafted units and nonpolar IPP chains might have an additional effect in the expansions at those low contents. Seemingly, this effect was more prominent in the dimension a. As the content of grafted units increased, on the other hand, the developed interactions between the polar side groups probably led to the decreases in lateral dimensions of the unit cells. The presence of the poly(BPOCPA) homopolymer molecules in the coproducts; however, resulted in the relatively lower contractions in the dimensions. Apparently, the interactions between the polar side groups of grafted and ungrafted poly(BPOCPA)

molecules might play role in the lower contractions of the lateral dimensions comparing to the copolymers. Moreover, the graft copolymerization resulted in the decreases in the parameter c; however, at all contents in both classes of the products, in consistence with the percentage, as depicted in Figure 8(c). The voluminous and polar side groups of grafted poly(BPOCPA) units probably brought about in the decreases in the dimension of the unit cells.

In addition to the unit cell parameters of the samples, the crystal size of copolymers and coproducts at different poly(BPOCPA) content were studied meticulously and the obtained result were drawn graphically in Figure 8(d). It was apparent from the figure that the crystal size increased initially with the percentage of poly(BPOCPA) in both classes of the products. The maximum crystal size in the coproducts, 23.77 nm (34.9% increase compared to the crystal size of neat IPP crystals) was observed at 7.3% poly(BPOCPA), while the maximum in the copolymers, 22.45 nm (27.4% increase) was observed at 7.6% poly(BPOCPA). Then, the maxima were followed by decreases with the content, gradually in the coproducts, but sharply in the copolymers. Nevertheless, despite of the decreases, the crystal size of all products were still

higher with respect to the neat IPP. It could be concluded that the poly(BPOCPA) units with regular structure (with the potential of forming the glassy nematic phase²⁵) might have acted as nucleating agents and promoted the growth of the crystals. Because, more ordered structures may lead to the formation of larger sizes of the crystals. The decreases after the maxima may be owing to the extensive nucleation at higher poly(BPOCPA) content, which may eventually result in a large number of IPP crystals, but with a smaller size.

3.4 | Mechanical properties of the products

In this present study, the mechanical crucial characteristic features of the products were also investigated by means of both tensile and impact tests performed at room temperature to determine the effect the content levels of poly(BPOCPA) depending on the being grafted or ungrafted units in the polymer matrices. The test samples (50 mm x 76.6 mm x 2 mm) were prepared by microinjection molding at 220°C with 8 bar injecting pressure. The processing was legitimately optimized at this fairly high temperature, comparing to the melting point of IPP, due to the difficulties in flow behavior of the melts at lower temperatures. The graft units poly(BPOCPA) with rigid and polar side groups might have caused to a restriction in chain mobility in the products, leading to an increase in the viscosity of the melts in the processing. Nevertheless, this processing temperature is well below the starting point of decomposition (245–250° as announced previously in Reference 21) of the graft units.

Typical stress versus strain curves belonging to the poly(BPOCPA)-g-IPP copolymers and the coproducts were presented in Figure 9. The extensive cold drawing

observed in only virgin IPP was not observed in any of the products even at low contents of the graft units. Moreover, it was apparent from the figure that, at low percentages, while the copolymer samples failed just after the yielding Figure 9(a), at the onset of neck formation, the samples of the coproducts failed at yielding as seen in Figure 9(b). At the percentages of 6.2% in the copolymers and 12.3% in the coproducts, the failures were recorded at the beginning of plastic deformation. At further respective contents, the products then displayed elastic deformation and brittle fracture in the tests. Apparently, the rigid character of poly(BPOCPA) molecules at room temperature (with T_g value of 69°C²¹), stemming from its large and rigid side groups with polar nature, restricts the plasticity behavior in the products. Conclusively, the material revealed an increasing brittleness in nature with increasing graft contents. Correspondingly, it can be deduced from the figure that the percent elongations belonging to the copolymer were relatively larger than that the coproducts had when compared the products at the same percentage of BPOCPA presenting in the reaction mixtures. It was directly attributed to the enhancement in the brittleness due to the rigid and stiff character of poly(BPOCPA). That is, the presence of comparatively high amount of poly(BPOCPA) polymers as grafted or ungrafted units in the products brought about the reduction in the ductile nature of the products since the additional hindrance effect caused by the restriction in IPP chain mobility came out in the polymer matrix.

In addition, the obtained results of the tensile and impact behavior measurements of the products were presented numerically in Table 3, and their variations were drawn in Figure 10. Tensile strength increased initially with the content in the copolymers, and reached the maximum 35.39 MPa at 6.2% poly(BPOCPA) (11.6% improvement comparing to pure IPP). The strength then

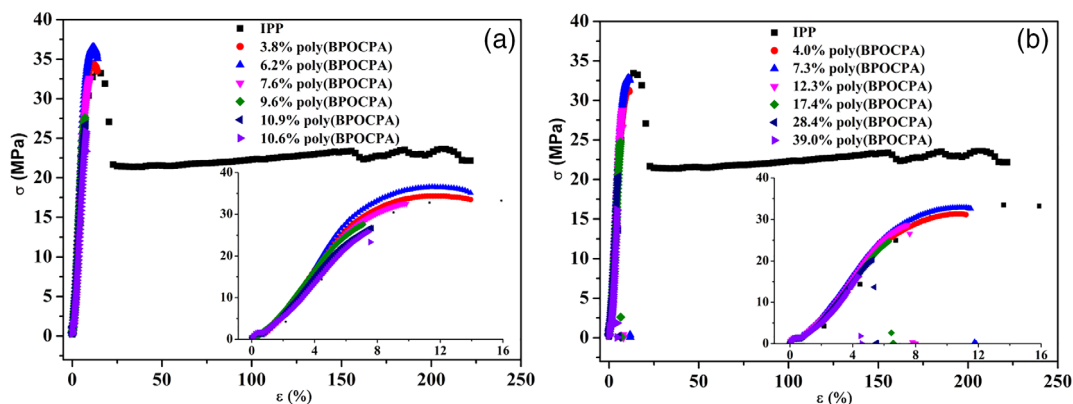
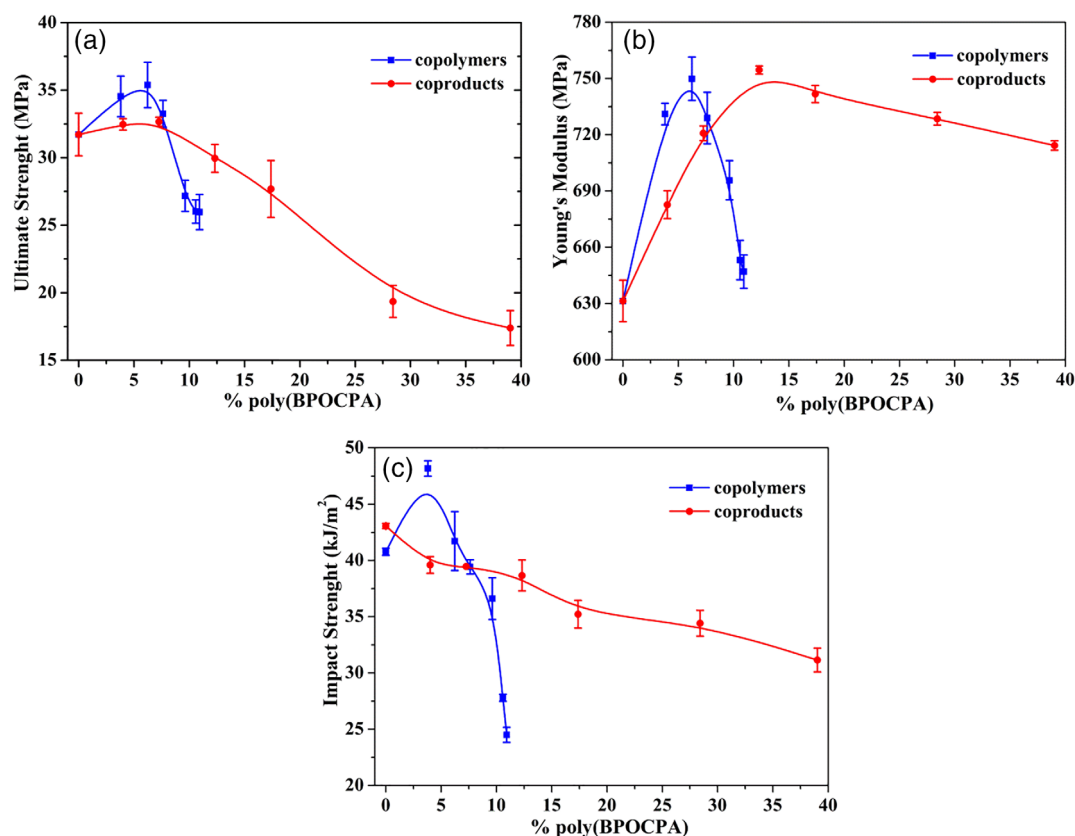


FIGURE 9 Stress versus strain curves of (a) the copolymers and (b) the coproducts with varying percentage of poly(BPOCPA). IPP, isotactic polypropylene; poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate) [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 3 Change of ultimate tensile strength, Young's modulus and impact strength of the copolymer and coproducts with poly(BPOCPA) content

Samples	Ultimate strength (MPa)	Young's modulus (MPa)	Impact strength (kJ/m ²)
Neat IPP	31.72 ± 1.57	631.35 ± 11.08	40.76 ± 0.31
3.8% poly(BPOCPA)	34.54 ± 1.51	730.99 ± 5.79	48.18 ± 0.68
6.2% poly(BPOCPA)	35.39 ± 1.68	749.84 ± 11.56	41.72 ± 2.61
7.6% poly(BPOCPA)	33.25 ± 0.10	728.91 ± 13.73	39.42 ± 0.63
9.6% poly(BPOCPA)	27.17 ± 1.15	695.70 ± 10.46	36.60 ± 1.86
10.9% poly(BPOCPA)	25.97 ± 0.86	646.96 ± 8.89	24.50 ± 0.68
10.6% poly(BPOCPA)	26.00 ± 1.31	653.12 ± 10.54	27.79 ± 0.31
4.0% poly(BPOCPA)	32.46 ± 1.57	682.67 ± 11.08	36.13 ± 0.98
7.3% poly(BPOCPA)	32.66 ± 0.34	720.79 ± 7.41	35.95 ± 0.01
12.3% poly(BPOCPA)	29.95 ± 1.03	754.52 ± 3.89	34.88 ± 1.83
17.4% poly(BPOCPA)	27.69 ± 2.10	728.53 ± 2.16	30.30 ± 1.63
28.4% poly(BPOCPA)	19.36 ± 1.18	728.53 ± 3.37	29.22 ± 1.53
39.0% poly(BPOCPA)	17.38 ± 1.29	714.22 ± 2.52	24.86 ± 1.41

Abbreviations: IPP, isotactic polypropylene; poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate).

**FIGURE 10** The changes in the (a) ultimate strength, (b) Young's modulus, and (c) impact strength of the copolymer and coproducts with different percentage of poly(BPOCPA). poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate) [Color figure can be viewed at wileyonlinelibrary.com]

weakened dramatically as the content increased, and reduced to 25.97 MPa at 10.9% poly(BPOCPA). Herein, it can be emphasized that the degree of crystallinity played

a dominant role in the mechanical performance of the copolymers besides grafting. Accordingly, one can see that there existed the apparent correlation between them

for the copolymers. It found that, the relatively larger degree of crystallinity led to the enhancement of the mechanical performance for the copolymer and vice versa. In the coproducts, however, the initial advance in the strength was not observed. The highest values 32.46 MPa and 32.66 MPa were recorded at 4.0% and 7.3% poly(BPOCPA), respectively. The presence of the poly(BPOCPA) homopolymer molecules thus led to the strengths that lay on the values comparable with the strength of pure IPP, 31.72 MPa. As the content increased further, the strength weakened gradually in consistence with the percentage, and decreased to 17.38 MPa at 39.0% poly(BPOCPA) as observed in Figure 10(a). On the other hand, very significant improvements were recorded in Young's modulus values of the products, determined on the same samples during tensile tests. The modulus increased initially with the content in both copolymers and coproducts, and reached the maximum 749 MPa (18.9% increase) at 6.2% poly(BPOCPA) in copolymers and 754 MPa (19.7% increase) at 12.3% poly(BPOCPA) in coproducts. It diminished as the content increased further in both classes of the products, sharply in copolymers but gradually in coproducts. Nevertheless, in spite of the dramatic decreases, it was apparent from the figure that all the moduli lay on the values still higher comparing to pure IPP, Figure 10(b). The graft copolymerization thus gave rise to a reinforcing function in withstanding under load and in resistance to deformation. These improvements in the tensile behavior are believed to result from the developments in the ordering and orientation of IPP chains due to the constitution of side chain LCP poly(BPOCPA) molecules as graft units.³⁰ The poly(BPOCPA) units with the potential of forming so regularly organized structure as nematic arrangement probably conducted to a better ordering and orientation of IPP chains, and thus gave rise to improved tensile behaviors as mentioned above. Both the decreases in the tensile behaviors after the maxima could be explained by a low load capacity of poly(BPOCPA) molecules. Because, the polymer composed of the chains with so large and rigid side groups are very difficult to exhibit substantial withstanding under the load with respect to IPP. Consequently, relatively higher content of poly(BPOCPA) units presumably made the material less prone to resist in the tensile tests.

In addition to that, the variation in Izod impact strength as a function of poly(BPOCPA) content in the products were presented in Figure 10(c). As seen in the figure, grafted poly(BPOCPA) units led to improvement in the strength of the copolymers at relatively low contents. The maximum 48 kJ/m² was achieved with the samples involving 3.8% poly(BPOCPA). The strength then decreased severely as the percentage increased further. The coproducts, on the other hand, had decrease trend in the strength in all samples, starting from the initial percentages and in consistence with

the content. It is well-known the fact that impact property or toughness of materials is its resistance to fracture. Namely, the higher impact strength is observed due to better dissipation of the energy loaded during the test, and the more the energy is absorbed, the higher impact strength the material has. Furthermore, various mechanisms may occur in (pure) polypropylene, which can lead to energy dissipation and thus to a higher toughness, for example, yielding, crazing, and phase transitions. The impact toughness is often the deciding factor in material selection because impact test measures the ability of polymer to withstand the load imposed upon being struck by an object at high velocity, thus it is a measure of energy required to propagate a crack cross the specimen. With respect to these explanations, the higher impact strength in copolymers compared to coproducts may be related to better energy dissipation in copolymers, nucleating agent properties of grafted poly(BPOCPA) content and changes in crystallinity of products. These results are consistent with DSC and XRD analyses. The percentage crystallinities were higher, whereas crystal sizes were lower in the copolymers compared to coproducts. In the study carried out by Jose et al. the increase in impact properties of PP on the addition of HDPE was related to the crystallinity changes arising due to blending. It was reported that the addition of HDPE decreases the size of PP spherulites by acting as a nucleating agent and increases impact strength of PP. Crystallinity of HDPE was higher when compared to IPP and pure HDPE has higher nucleation and growth of individual crystallites.³¹ Moreover, a reduction in spherulite size was reported to improve impact strength due to an increase in interfacial thickness, better interspherulitic chain mobility, and reduction of spherulitic defects. In addition, fracture toughness in semi-crystalline polymers strongly depends on the number of tie molecules between the crystallinities and the disentanglement resistance of the polymer chains (tie molecules connect crystals and amorphous region).³² In the light of these explanations, at high content of poly(BPOCPA), the shrinkage of the basal area as founded in XRD results presumably gave rise to the diminishment in the free moving capability (degree of freedom) of IPP polymer chains in the matrix. As a consequence of this effect, the products possessed the relatively lower impact strength values since the generated energy (stress) by suddenly applied impact force did not delocalized efficiently and did not distributed uniformly throughout the IPP polymer matrix.

3.5 | Morphological properties of the products

It is known that size, size distribution of reinforcement phase and the adhesion quality of interface between

matrix and the reinforcement phase are important factors for the mechanical performance of polymer blends and composites.^{33–35} Thus, the morphological properties of the tensile and impact fractured surfaces of the copolymers and the coproducts were examined meticulously by SEM analysis in order to unfold how the presence of varying amount of poly(BPOCPA) in the polymer matrix affected the surface topographies of the products. The SEM images (taken with varying magnifications) regarding the tensile and impact fractured surfaces belonging to the copolymers and the coproducts were presented in Figures 11 and 12, respectively. One can clearly observe that, all the fracture surfaces displayed no phase separation. The structures of the products were completely

homogeneous in spite of that the graft units poly(BPOCPA) with polar side groups was essentially different in nature from a polar IPP. This result confirmed the chemical bonding of the units, that is, graft copolymerization of BPOCPA onto IPP. Furthermore, since poly(BPOCPA) existed also both as homopolymer as well as the grafted units in the coproducts, it was additionally indicated that the graft copolymers poly(BPOCPA)-g-IPP may act as compatibilizers providing good interfacial adhesion between the homopolymers and graft copolymers in the coproducts. Furthermore, the fractographs revealed that brittleness governed the fractures of the samples in general. Some extent of ductility was also detected at relatively low contents of poly(BPOCPA),

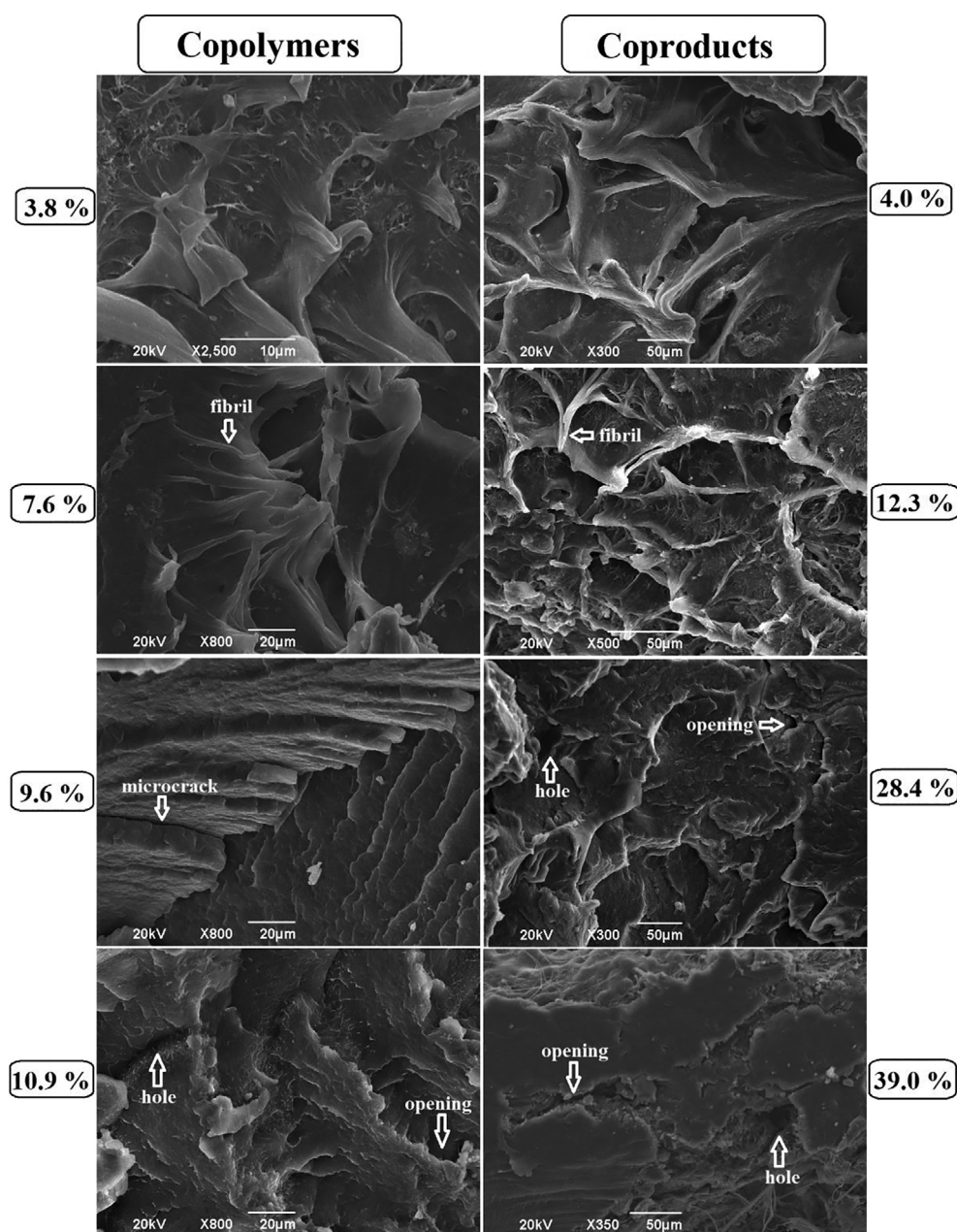
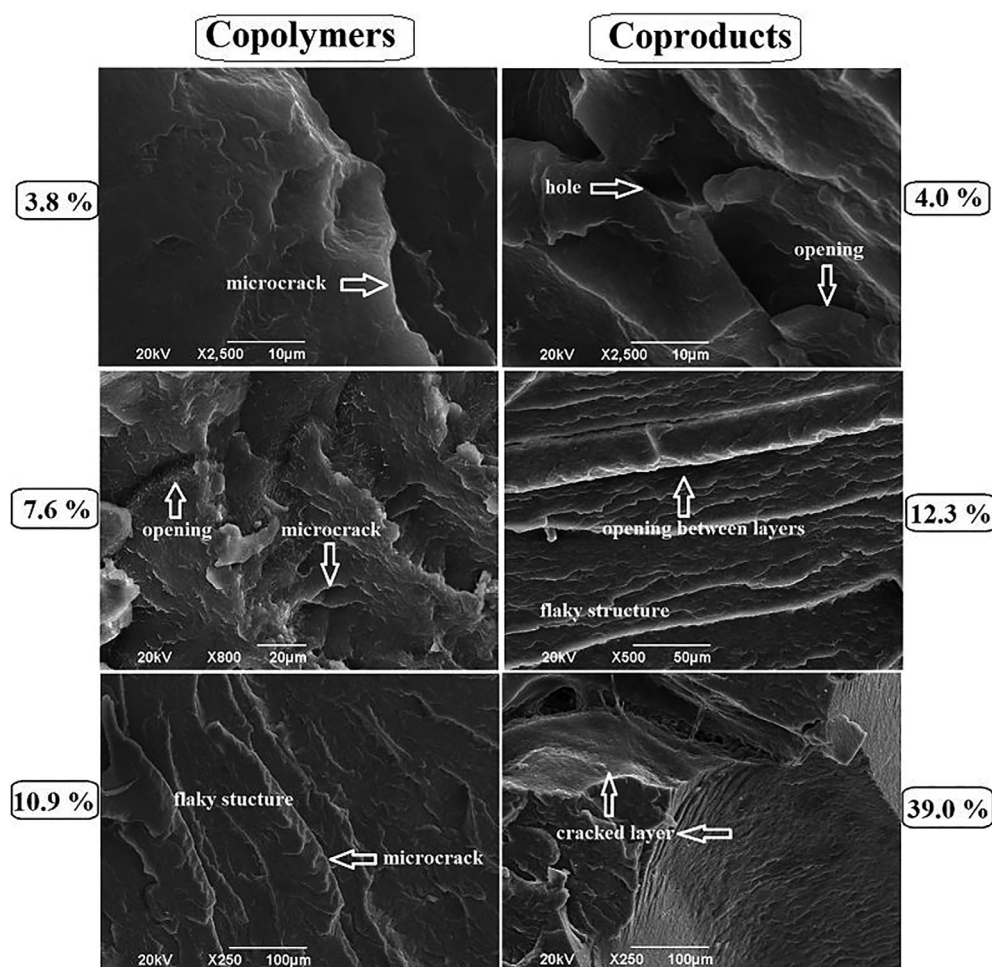


FIGURE 11 SEM micrographs of tensile fractured surfaces belonging to the copolymers and coproducts containing varying percentage of poly(BPOCPA). poly(BPOCPA), poly(p-benzophenoneoxycarbonylphenyl acrylate); SEM, scanning electron microscope

FIGURE 12 SEM micrographs of impact fractured surfaces belonging to the copolymers and coproducts containing varying percentage of poly(BPOCPA). poly(BPOCPA), poly(*p*-benzophenoneoxy-carbonylphenyl acrylate); SEM, scanning electron microscope



especially at 7.6% and 12.3% content levels. This was seen as a experimental sing regarding that the better improvements in the tensile behavior of the copolymers and coproducts were obtained thanks to the better alignment and orientation in the fibrillar structure. It was observed some short fibrillar structure broken in ductile besides the microcracks and openings showing the major fracture went through brittle nature in Figure 11. Unfortunately, the prominent change in the morphology was clearly seen through the fractographs at hight content, especially at 10.9% and 39.0% of poly(BPOCPA), at which the short fibrillar structure seen at low contents was replaced by cracks, openings and holes in the fractures. But, the photographs did not show any phase separation at all products as if the morphological properties became worse. These appearance defects explained why the decrements in the tensile strength and modulus values of the products were recorded. In addition to that, the examination of the fractured surfaces obtained from the impact tests were performed by means of SEM in this current study. The micrographs (SEM images) of the impact fractured surfaces of some copolymers and coproducts presented in Figure 12. The taken SEM images revealed that the

fractures were due to the explicitly brittle character of the products. It is known that, in impact testing, a force is loaded onto the certain area of the test samples in a small time scale. If the loaded impact energy is not delocalized effectively at molecular level, that is, by orientation of the macromolecules, the energy is localized in the weak regions of the structure to form cracks and voids. According to that, the formation of the zigzagged propagation of the cracks was observed in the products due to the noneffective distribution of the generated impact force throughout the polymer matrix as seen in Figure 12. Furthermore, at high content of poly(BPOCPA) in the products (especially in the coproducts), the obtained fractographs showed why the reduction in the resistance of the material against to the applied force was recorded due to presumably decrement in the degree of crystallinity³⁶ (See Figure 6). In other words, since the fracture toughness in semicrystalline polymers strongly depends on the number of tie molecules between the crystallites and disentanglement resistance of polymer chains as previously mentioned,³² the crystallization behavior of the products, meaning the percent crystallinity and crystal size, might also play role in

the observed mechanical behaviors. As a consequence of that, the holes, microcracks, openings among the layers, omnipresent flaws, holes, layered structure as well as the formation of the flaky layers and microstructural defects were observed in SEM images belonging to both classes of the products.

4 | CONCLUSIONS

Along the current article, the role of poly(BPOCPA) presenting not only as only grafted in the copolymers but also as grafted and ungrafted units in the coproducts on the fundamental characteristic features of IPP were investigated in details. The changes in crystalline melting temperature, degree of crystallinity, unit cell parameters, grain size as well as tensile strength, moduli and impact strength depending on the morphological properties were determined for both classes of the products with the aid of conventional experimental measurements techniques; namely DSC, XRD, universal mechanical testing, and SEM. Thanks to the obtained results from the current study, the novel and significant links about the effect of poly(BPOCPA) on IPP properties were established. The major empirical evidences to be deduced from this work were given as followings:

- Thermally initiated graft copolymerization of BPOCPA onto IPP was successfully accomplished. The changes in the amount of poly(BPOCPA) in the copolymers and the coproducts were investigated as a function of monomer percentage in the reaction mixtures. Poly(BPOCPA) percentage in all the products were increased with the percentage of the monomer in the reaction mixture. However, 5% of monomer content was found to be optimum content level in order to obtain the maximum percent grafting among the copolymers.
- The existence of relatively higher amount of poly(BPOCPA) polymers in the matrix led to remarkable enhancement in the melting temperature of the coproducts, but the variation in the melting temperatures in the copolymer samples were recorded. Moreover, the larger degree of the crystallinity were recorded with the copolymers when compared to the coproducts. However, at higher poly(BPOCPA) content, the crystal quality of the copolymers get worse and worse, while the coproducts get better in terms of crystallinities since the regularly organized poly(BPOCPA) units might have acted as nucleating agents in the polymer matrices.
- The lateral dimensions of the monoclinic unit cell of IPP matrix were also found to be considerably affected

by poly(BPOCPA) content in the products. Both a and b parameters increased significantly at low percentages of the poly(BPOCPA) due to the fact that IPP chains in the unit cells were probably forced apart laterally by the polar graft units, but these increment in the coproducts were larger than the copolymers due to the existence of the weak interactions between polar side groups of the grafted units and nonpolar IPP chains. At higher contents, the results depicted that the enhancement in the interactions between the polar side groups presumably gave rise to the decrement in lateral dimensions of the unit cells. The parameter c diminished steadily in all products. Moreover, it was concluded that the presence of lower amount of poly(BPOCPA) units led to the advance in the crystal sizes in the both classes of the products, but the further increment in the percentage of poly(BPOCPA) created negative effect in terms of crystal size of the products.

- As for the results deduced from the mechanical tests, the graft copolymerization of BPOCPA onto IPP gave rise to some improvements in tensile strength and modulus of the products at lower graft content and it was founded that the mechanical performance of the copolymers were greater than the coproducts. On the other hand, the impact strength of the copolymer enhanced at low content, which was followed by dramatic declining at high content. The existence of comparatively higher percent of poly(BPOCPA) made the polymer worse and damage seriously due to the increasing in the brittleness and the reduction in the free moving capability (degree of freedom) of IPP polymer chains in the matrix.
- With respect to SEM micrographs, no phase separation was observed at all products, although the graft units poly(BPOCPA) with polar side groups were essentially different in nature from a polar IPP. This can be seem a sign regarding the formation of the chemical bonding between BPOCPA and IPP. The ductile characteristic appearance were observed at low percentage of poly(BPOCP) in all the products, the brittle nature became more dominant at the further content levels.

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